



Bahria University, Islamabad

Department of Software Engineering

Machine Learning Lab

(Fall-2026)

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Lab Journal: 01

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Task No:	Task Wise Marks		Documentation Marks		Total Marks (20)
	Assigned	Obtained	Assigned	Obtained	
1	3		5		
2	3				
3	3				
4	3				
5	3				

Comments:

Signature

Title: Study of Python Libraries for ML application such as Pandas and Matplotlib.

Objective:

- To understand data analysis using Pandas library
- To understand data visualization in the form of 2D graphs and plots using Matplotlib library

```
[18] ✓ Os
# Load the dataset from your uploaded local file
df = pd.read_csv('tips.csv')

print("Dataset loaded successfully!")

Dataset loaded successfully!

[19] ✓ Os
# Display the first 5 rows
df.head()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
[20] ✓ Os
# Count missing values per column
df.isnull().sum()

0
total_bill 0
tip 0
sex 0
smoker 0
day 0
time 0
size 0

dtype: int64
```

```
[21]
✓ Os
# Generate summary statistics for numeric columns
df.describe()
```

	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
25%	13.347500	2.000000	2.000000
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000

```
[22]
✓ Os
# Calculate median values
df.median(numeric_only=True)
```

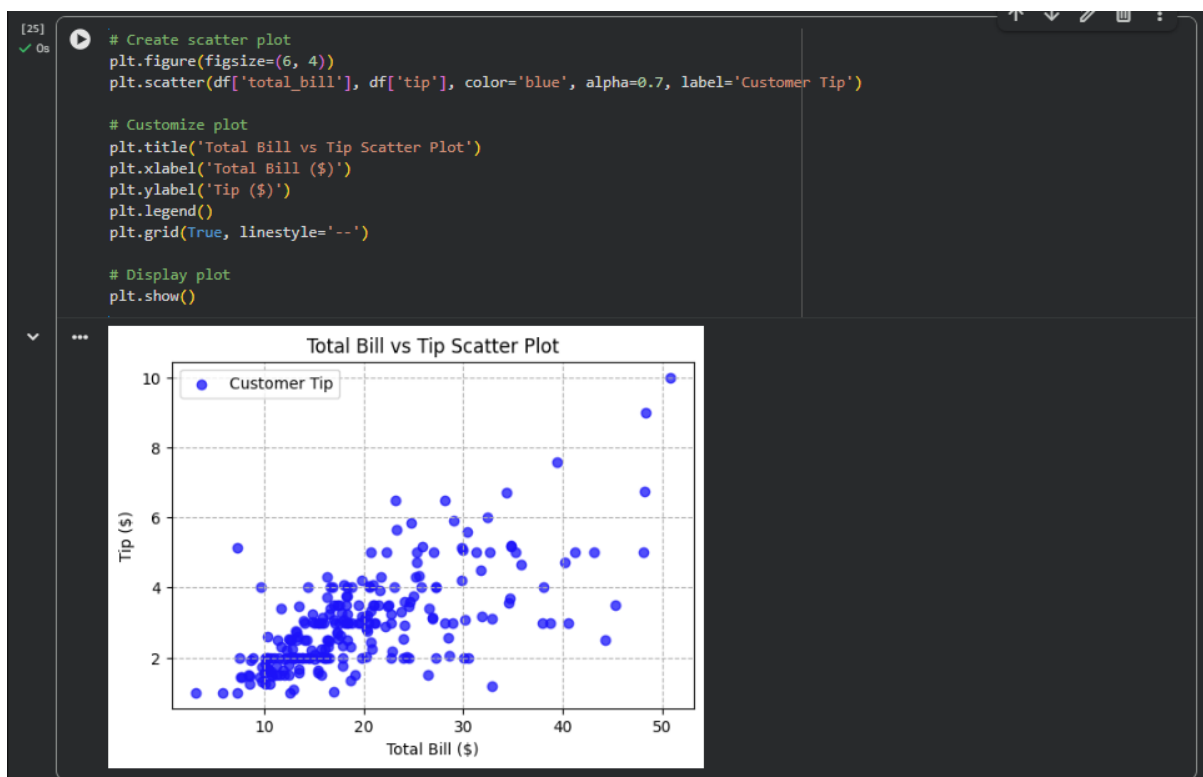
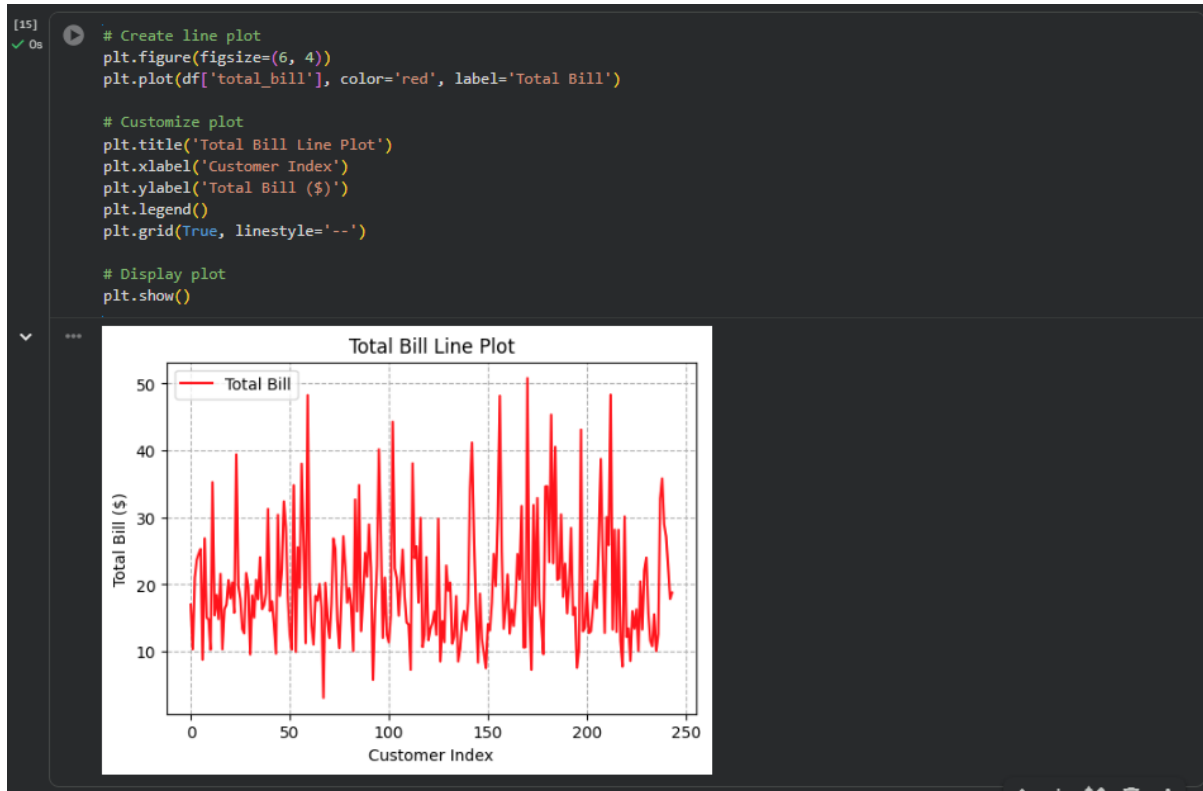
total_bill	17.795
tip	2.900
size	2.000

dtype: float64

```
[23]
✓ Os
# Filter rows where total_bill > 20
high_bills = df[df['total_bill'] > 20]

# Display the first few rows of the filtered subset
high_bills.head()
```

	total_bill	tip	sex	smoker	day	time	size
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
5	25.29	4.71	Male	No	Sun	Dinner	4
7	26.88	3.12	Male	No	Sun	Dinner	4





Viva task:

```
[30] ✓ 0s df[df['day'] == 'Sun']
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
...	...	...	...	...	...	...	...
186	20.90	3.50	Female	Yes	Sun	Dinner	3
187	30.46	2.00	Male	Yes	Sun	Dinner	5
188	18.15	3.50	Female	Yes	Sun	Dinner	3
189	23.10	4.00	Male	Yes	Sun	Dinner	3
190	15.69	1.50	Male	Yes	Sun	Dinner	2

76 rows × 7 columns

Thank you
